

readxl (1)

INTRODUCTION TO IMPORTING DATA IN R



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Microsoft Excel

- Common data analysis tool
- Many R packages to interact with Excel
- readxl - Hadley Wickham

Typical Structure Excel Data

- Different sheets with tabular data

Capital	Population		
New York	16044000		
Berlin	3433695		
Madrid	3010492		
Stockholm	1683713		
year_1990			Population
			17800000
			3382169
			2938723
year_2000			1942362



readxl

- `excel_sheets()`
 - list different sheets
- `read_excel()`
 - actually import data into R

```
install.packages("readxl")  
library(readxl)
```

excel_sheets()

```
dir()
```

```
"cities.xlsx" "the_rest_is_secret.txt"
```

```
excel_sheets("cities.xlsx")
```

```
"year_1990" "year_2000"
```

read_excel()

```
read_excel("cities.xlsx")
```

```
# A tibble: 4 × 2
  Capital Population
  <chr>      <dbl>
1 New York  16044000
2 Berlin   3433695
3 Madrid   3010492
4 Stockholm 1683713
```

```
read_excel("cities.xlsx", sheet = 2)
read_excel("cities.xlsx", sheet = "year_2000")
```

```
# A tibble: 4 × 2
  Capital Population
  <chr>      <dbl>
1 New York  17800000
2 Berlin   3382169
3 Madrid   2938723
4 Stockholm 1942362
```

Let's practice!

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readxl (2)

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read_excel()

```
read_excel(path, sheet = 1,  
           col_names = TRUE,  
           col_types = NULL,  
           skip = 0)
```

Capital	Population		
New York	16044000		
Berlin	3433695		
Madrid	3010492		
Stockholm	1683713		
year_1990			
year_2000			



read_excel() - col_names

```
read_excel(path, sheet = 1,  
           col_names = TRUE,  
           col_types = NULL,  
           skip = 0)
```

- `col_names = FALSE`: R assigns names itself
- `col_names = character vector`: manually specify

read_excel() - col_types

```
read_excel(path, sheet = 1,  
           col_names = TRUE,  
           col_types = NULL,  
           skip = 0)
```

```
read_excel("cities.xlsx", col_types = c("text", "text"))
```

```
# A tibble: 4 × 2  
  Capital Population  
  <chr>      <chr>  
1 New York  16044000  
2 Berlin    3433695  
3 Madrid    3010492  
4 Stockholm 1683713
```

read_excel() - col_types

```
read_excel(path, sheet = 1,  
           col_names = TRUE,  
           col_types = NULL,  
           skip = 0)`
```

```
read_excel("cities.xlsx",  
           col_types = c("text", "blank"))
```

```
# A tibble: 4 × 1  
  Capital  
  <chr>  
1 New York  
2 Berlin  
3 Madrid  
4 Stockholm
```

read_excel() - skip

```
read_excel(path, sheet = 1,  
           col_names = TRUE,  
           col_types = NULL,  
           skip = 0)
```

```
read_excel("cities.xlsx",  
           col_names = c("Capital", "Population"),  
           skip = 2)
```

```
# A tibble: 3 × 2  
  Capital Population  
  <chr>      <dbl>  
1   Berlin    3433695  
2   Madrid    3010492  
3 Stockholm   1683713
```

Wrap-up

- `excel_sheets()`
- `read_excel()`
- Everything you need!
- Fast
- Same arguments as in `readr` package
- Consistency

Let's practice!

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